From Alerts to Certified Rules: Self-Consolidating Intrusion Detection for Power Grids

Draft for IEEE Transactions on Smart Grid (regular paper). Every number is copied from a generated report in `paper3-wp0/reports/`*, named in square brackets. Bibliographic details of references [4], [9], [10], [13], [14], [19], [21] and [23] were checked against publisher or index records; the remaining classic references should be checked once more against the originals before submission.*

ABSTRACT

Anomaly detectors on power-grid telemetry raise alerts, but each alert is consumed and forgotten: nothing durable, inspectable or auditable remains, and the next similar attack is detected, or missed, from scratch. Hand-written detection rules are durable but brittle, because benign operating changes (load variation, reconfiguration, tap changes, maintenance) trigger them. We propose a self-consolidating intrusion detection system that turns every alert into deterministic detection rules and admits a rule into a rule library only after a **physics oracle** has certified that no benign, physically feasible operating state triggers it. The oracle has a search layer (Latin hypercube sampling and differential evolution over AC power flows) and an exact layer (an SMT solver over a linearised AC model, made robust by adversarially validated model-error bounds); counterexamples refine the rules. The library is then replayed without labels: rules learned from earlier alerts catch later attacks, sometimes attacks the detectors miss, and sometimes earlier. We evaluate on the public Sherlock dataset, on three distribution grids reconstructed from data and validated at the measurement noise floor: a fully observed 15-bus CIGRE MV grid, a 178-bus SimBench semi-urban grid with about 40 % of elements unmonitored, and a 99-bus SimBench rural grid used as a zero-training transfer target. Pooled over 103 process-attack instances on the recorded test days and an additional, independently calibrated test day, combining the certified library with a classic physics detector detects 48 % of instances against 34 % for the detector alone, with 14 instances detected only by the combination and none only by the detector (exact McNemar test, $p = 0.00012$). On the partially observed grid, the library detects 11 of 21 injected attacks at zero false alarms where state-estimation bad-data detection catches 2. We also report what did not work: rules mined from history are refined into uselessness, machine-learning detectors on raw telemetry raise alarms on 43 % to 99.99 % of benign records, and per-measurement margins need per-grid and per-topology noise calibration, which we provide online and without labels.

Index Terms: intrusion detection, cyber-physical security, power distribution grids, SMT solving, counterexample-guided synthesis, state estimation, SCADA.

I. INTRODUCTION

The control centre of a distribution grid receives thousands of measurements per second over legacy protocols such as IEC 60870-5-104, and sends commands over the same channels. The 2015 and 2016 attacks on the Ukrainian grid [1], and the more than 200 cyber incidents recorded against the German energy sector between 2023 and 2024 [2], [14], show two attacker options: falsify what the operator sees, or issue valid commands that

change the physical grid. Industrial intrusion detection systems (IIDS) are the accepted retrofittable defence [3], [4]. Two families dominate. **Anomaly detectors**, physics-based or learned, compare telemetry with a model of normal behaviour and raise alerts. **Rule-based detectors** encode what an operator knows should never happen. Each family has a well-known weakness. Alerts leave no durable knowledge: the detector that flagged a frozen RTU today has no memory of it tomorrow, and an analyst cannot inspect why it fired. Rules are durable and inspectable, but every benign operating change that their author did not foresee becomes a false alarm, and in a distribution grid such changes are the norm: loads follow profiles, DER output follows weather, operators reconfigure feeders, change transformer taps and take lines out for maintenance.

This paper proposes to close the gap between the two families with a **certification gate**. An anomaly engine runs continuously. Each time it raises an alert that no rule in the library covers, the system synthesises candidate rules from the telemetry around the alert. A candidate enters the library only after a **physics oracle** has searched for a benign, physically feasible operating state that would trigger it, that is, a false alarm, and found none. Counterexamples found by the oracle are returned to the synthesiser, which refines the rule (counterexample-guided inductive synthesis, CEGIS [5]). The result is a library of deterministic rules, each with a certificate that says “no state inside the benign operating envelope of this grid fires this rule, up to a validated model error”. The library grows with every incident, is inspectable rule by rule, and detects later attacks on its own.

The idea is simple to state and hard to make work. The oracle needs a grid model (the dataset contains none; all three were reconstructed from telemetry), a definition of “benign and feasible” an operator would sign, a search that reaches the corners of a high-dimensional envelope with hundreds of topologies, an exact layer that proves absence, and a noise model that keeps a certified rule from firing on noise. The contributions are:

1. **A two-layer physics oracle for detection rules.** A search layer over the full AC power flow and an exact layer that encodes envelope, linearised AC power flow and rule in linear real arithmetic for an SMT solver [6]. The exact layer is made *robust*: each AC condition is replaced by the weakest linear condition it implies under model-error bounds that are themselves validated by adversarial search, and the bounds carry a probabilistic guarantee from the scenario approach [7]. The linearised AC model covers meshed configurations, which LinDistFlow [8] does not, and makes power-balance rules certifiable.
2. **Rule synthesis from alerts** in three families, physical invariants (one measurement against its linear prediction from metered injections), grid-wide invariants per topology, and state rules (a discrete change not present in the benign envelope), with counterexample-guided refinement of margins. We publish the synthesised rules (Section VI-B).
3. **A self-consolidation loop** evaluated without labels: alerts come only from detectors, rules act only after they are learned, labels are used afterwards for scoring. We report a strict variant that credits a rule only with attacks that started after it was learned.
4. **Partial observability and unseen topologies.** For a 178-bus grid with about 40 % of elements unmonitored, the invariant becomes a state-estimation residual with pseudo-measurements and a Gauss-Newton correction against the full AC model, and each new topology is certified on demand in 5 to 11 s.
5. **Evaluation discipline.** Three reconstructed grids validated against ground truth, a zero-training transfer experiment, an additional independently calibrated test day, confidence intervals and paired tests, seed and parameter ablations, and the negative results that shaped the design.

Section II reviews related work, Section III states the problem and threat model, Section IV the method, Section V the experimental setup, Section VI the results, Section VII the discussion and limitations, and Section VIII concludes.

II. RELATED WORK

Process-aware intrusion detection. Network-level IIDS inspect protocol fields and traffic patterns, and miss attacks that are syntactically valid [3]. Process-aware IIDS model the physical process state carried by the protocol. Hadžiosmanović et al. [9] mine value ranges and update patterns per variable from PLC traffic; Feng et al. [10] derive invariants for an industrial process from data with a systematic framework; Chen et al. [11] learn invariants with code mutation and Lin et al. [12] with graphical models and Bayesian networks. IPAL [13] gives a protocol-independent process-state representation on which such detectors can run; the Sherlock dataset [14] uses it and evaluates five recent IIDS, finding that none handles all of its attack types. Our work differs in two ways. The rules are not mined from history and trusted; they are synthesised from alerts and certified against the physics of the grid, and we show that a rule mined from history alone is refined into uselessness when made safe (Section VI-A). And the rule library is the output of an online loop fed by anomaly detectors, not a one-time specification.

Physics-based attack detection and bad-data detection. The residual between measurements and a physical model is the classic detection statistic, in state estimation as bad-data detection (BDD) [15] and in control systems as residual tests with persistence (CUSUM-style) [16], [17]. Liu, Ning and Reiter [18] showed that an attacker who knows the measurement model can craft false data that passes BDD, which spawned a large literature on stealthy false-data injection and its detection, surveyed in [19]. Our detectors D1 (full power flow) and SE-BDD (weighted least squares with a Gauss-Newton correction) are instances of this family and serve as the alert sources and as the classic baseline. The certified library differs from BDD in granularity: BDD has one global threshold on the largest normalised residual, whereas a certified rule has one margin per measurement, calibrated to that measurement's noise and to the oracle's bound for that topology. On the partially observed grid this difference accounts for the library's advantage (Section VI-E).

Formal methods for rules and controllers. SMT solvers [6] and CEGIS [5] are standard tools for synthesising programs and controllers from specifications and counterexamples. In power systems they have been used for verification of protection logic and of controller invariants, mostly with DC or linearised models. Our contribution is to use them as a *certification oracle for detection rules*, with a sound treatment of the linearisation error: the exact layer answers "no benign counterexample" only up to a model-error bound that we validate adversarially and bound probabilistically with the scenario approach of Calafiore and Campi [7].

Machine learning for grid intrusion detection. Autoencoders, isolation forests [20] and graph neural networks are widely proposed for anomaly detection on grid telemetry. The Sherlock dataset paper [14] reports that recent systems struggle with benign operating change. We confirm this on three grids: raw-feature detectors raise alarms on most benign records (Section VI-H), and physics-informed features fix it only where the training day covers the topologies of the test day. We use ML detectors as alert sources and baselines, not as the final decision.

Datasets. Sherlock [14], generated with the Wattson co-simulator [21] on CIGRE [22] and SimBench [23] grids, is the only public dataset we know of with recorded benign operator procedures (maintenance, ring closing, tap changes) interleaved with attacks on a distribution grid, which is exactly what a certification gate must survive. Its download contains no grid model, so we reconstruct the grids with pandapower [24] (Section V-B).

III. PROBLEM STATEMENT AND THREAT MODEL

System. A distribution grid with buses $\mathcal{B}$, lines $\mathcal{L}$, transformers, loads, DER units and storage, operated from a control centre that receives a telemetry vector $m \in \mathbb{R}^n$ at 1 Hz: bus voltage magnitudes, line and transformer active-power flows, currents and loadings, injections of loads, DER and storage, switch states and tap positions. In the partially observed grid a fixed subset of elements has no telemetry.

Benign envelope. The set $\mathcal{E}$ of operating states an operator considers normal: load injections inside per-element ranges, controllable DER and storage over their physical ranges, load reactive power over its observed range, transformer taps in a set of positions, and a set of topologies (switch configurations) that keep the grid connected, with operating limits (bus voltage in $[0.90, 1.10]$ pu, line and transformer loading at most 100 %). A state is benign if it lies in $\mathcal{E}$ and its AC power flow converges within the limits.

Attacker. The attacker controls the communication path between RTUs and the control centre, or an RTU itself, and can (a) falsify reported measurements (drift a voltage reading, freeze a set of readings at stale values), and (b) send valid commands that change the physical grid (open a line, disconnect a generator or a load, change a set point). Attack (a) makes the telemetry physically inconsistent; attack (b) keeps it consistent but may create a state outside $\mathcal{E}$. We do not assume knowledge of the attack types at design time: rules are synthesised from whatever the detectors flag.

Rule. A rule ρ is a Boolean function of m, written in a small language (Section IV-B). It fires on state m if $\rho(m)$ holds. A rule is **certified** if no benign state fires it: $\forall s \in \mathcal{E}: \neg\rho(m(s))$, where $m(s)$ is the telemetry the AC power flow produces for s. Measurement noise is treated separately: a certified rule is deployed with a margin at least the noise level observed on the grid (Section IV-E).

Goal. Maintain a library $\mathcal{R}$ of certified rules, grown from detector alerts without labels, such that later attacks are detected by $\mathcal{R}$ with a false-alarm rate on benign records comparable to the detectors', and such that each rule can be read and audited.

IV. METHOD

A. Overview

Fig. 1 shows the loop. (1) Detectors watch the telemetry. (2) An alert opens a 60 s window. (3) If a rule in the library already fires within the window, the alert is covered and nothing is synthesised. (4) Otherwise candidate rules are synthesised from the window and the topology at the alert. (5) The physics oracle certifies or refutes each candidate; counterexamples refine margins. (6) Certified rules enter the library and act from the time they were learned.



Fig. 1

Fig. 1. End-to-end architecture. Telemetry feeds detectors; an uncovered alert opens a 60 s window from which candidate rules are synthesised; the two-layer physics oracle certifies or refutes them, with counterexamples refining margins; certified rules enter the library, which is matched against live telemetry after a short online noise calibration per new topology. Labels are never used inside the loop.

B. Rule language

A rule is a disjunction of conjunctions of atoms. Three atom types: a **threshold** on one measurement ($m_i < c$ or $m_i > c$), a **linear** inequality $\sum_k a_k m_k \lessgtr c$, and a **discrete** equality on a switch, line or tap state. Each atom has a scale that normalises units. The **margin** of a rule on a state is the maximum over clauses of the minimum over atoms of the atom margin ((c − lhs)/scale for $<$, the negative for $>$, ±1 for discrete atoms); the rule fires when the margin is positive. Margins give the search layer a continuous objective and make "how close to firing" comparable across rules. Rules are stored as JSON and are human-readable; Section VI-B lists the ones the loop produced.

C. Benign envelope

The envelope is explicit and documented per grid. Continuous parameters: per-load active power as a fraction of nominal, within the range observed on the attack-free training day widened by 0.01; DER output in $[0, 1]$ of rated and storage in $[-1, 1]$ of nominal (operators change set points, and the test days contain benign set-point changes not seen in training); load reactive power either fixed at nominal or on its own observed range, whichever the data show. Discrete parameters: transformer tap positions seen in training (optionally padded by one step, Section VI-F), and topologies. For the 15-bus grid the topologies are enumerated: every switch configuration that keeps the grid connected with at most one closed loop and up to two lines out, 336 configurations (170 radial, 166 with one loop) [sherlock_envelope_sherlock.md]. For the 178-bus grid the topology space is structural (lines in or out of service) and each topology is certified when it first appears (Section IV-H). Benign operating limits are as in Section III.

D. The physics oracle

Search layer. Stage A draws 10,000 Latin hypercube samples of the continuous parameters across configurations, solves each AC power flow, keeps the benign ones and records whether any fires the rule. Stage B runs differential evolution [25] on the rule margin, per configuration or jointly over the configuration index when there are many, with a fixed power-flow budget (32,000 per rule on the 15-bus grid). Power flows use a Newton-Raphson solver built on pandapower's admittance matrices, dense for small grids and sparse above 60 buses; it agrees with pandapower's `runpp` to 5.0×10^{-7} over all measurements and runs in 2.2 ms against 35.2 ms per flow [envelope_smoke.md]. The search layer gives a **refutation** when it finds a benign firing state (re-checked with `runpp`); it never proves absence.

Exact layer. Z3 [6] over linear real arithmetic with three parts: the envelope box and the configuration, a linear power flow model, and the rule. Two linear models are used. LinDistFlow [8] is radial only and ignores losses. The **linearised AC** model is the first-order expansion of the AC power flow at the envelope centre of each configuration; it covers meshed configurations, and its worst-case voltage error found by adversarial search is 0.00708 pu on the reference grid and 0.00834 pu on the 15-bus Sherlock grid, against 0.01204 and 0.01207 pu for LinDistFlow [linac_error.*md, lindistflow_error*.md]. UNSAT in the exact layer is a certificate *relative to the linear model*.

Robust encoding. Each AC condition is replaced by the weakest linear condition it implies under a model-error bound δ_q per quantity q: a rule atom $m_q > c$ becomes $\hat{m}_q > c - \delta_q$, and likewise for operating limits. UNSAT then means: no benign counterexample exists *provided the linear-to-AC error stays within* δ. The δ_q are 1.25 times the larger of the sample maximum and the worst case found by differential evolution on the error itself (500 power flows per topology and quantity) [linac_error_sherlock.md]. For linear atoms over several measurements a rule-specific bound on the atom expression is tighter than the sum of per-quantity bounds (0.287 against 0.538

MW for the power-balance rule [atom_bounds_linac_sherlock.md]). Box-implied constraints are pruned before solving; queries run in parallel over configurations. A SAT model counts as a refutation only when re-checked as a benign AC counterexample; UNSAT in the robust encoding is a certificate; otherwise the search verdict stands.

Probabilistic bound on the model error. The adversarial search gives an empirical worst case, not a proof. We add a distribution-based guarantee [scenario_bound.md]: for N = 20,000 independent benign states drawn uniformly over the envelope, every δ_q exceeds the sample maximum of the error, and by the scenario approach [7] a new benign state drawn the same way exceeds δ_q with probability at most $\epsilon = 1 - \beta^{1/N} = 3.45 \times 10^{-4}$ at confidence $1 - \beta = 0.999$. This bound is relative to the sampling distribution and is stated as such. Fig. 2 shows the three quantities per measured kind: the sample maximum, the adversarial worst case and the bound δ the exact layer uses.

 Fig. 2

Fig. 2

Fig. 2. Validated model-error bounds on the 01-Basic grid: for each quantity, the largest linearised-AC error over 1,000 benign samples, the worst case found by differential evolution, and the bound δ = 1.25 × worst that the robust exact layer uses [linac_error_sherlock.md].

Counterexample-guided refinement. On a counterexample a threshold rule is tightened just past the counterexample value, and an invariant rule's margin is raised to the synthesis rule below with the updated oracle bound, for at most five rounds. The refinement trace is reported together with how much of the rule's original firing region remains, so that a rule refined into uselessness is visible (Section VI-A).

E. Rule synthesis from alerts

Physical invariant (family P). For the topology T at the alert and a reported quantity y (a bus voltage or a line active-power flow), the linearised AC model gives $y \approx c_T + a_T^\top x$ where x are the metered injections. The rule is $|y - c_T - a_T^\top x| > \mu$ with a discrete guard that the topology is T. The margin is

$$\mu = 1.5 \times \max(\text{noise}, \text{bound}),$$

where *noise* is the largest invariant residual over the attack-free training records in T (floored by the largest residual over all training records when T has fewer than 300 training records), and *bound* is the largest residual the oracle finds over benign AC states of T, times the 1.25 safety factor. The factor 1.5 was fixed before any result and must exceed the 1.25 bound safety factor (Section VI-G). Candidate quantities are those whose residual exceeds the noise in the alert window.

Grid-wide invariant (family G). One rule per topology with one clause per bus voltage and per active line flow, each the P invariant above in both signs. It fires when any monitored network quantity is inconsistent with the metered injections in that topology. One G rule covers every later falsification in that topology, so the loop synthesises it once per topology.

State rule (family S). A discrete change observed between the 30 s before the alert and the alert window: lines going out of service, loads, DER or storage going to zero. The candidate is the conjunction of these observations (thresholds at half the pre-alert value for injections). The oracle decides whether such a state exists in the benign envelope: an isolated zone with its loads at zero does not, a single DER at zero output does.

State-estimation residual rule (family SE), partial observability. Section IV-H.

F. Self-consolidation loop and evaluation protocol

The test day is replayed in time order. Alerts come from detectors only. Alarm runs closer than 120 s form one episode. For each episode, in order: if the library fires within the first 60 s, the episode is covered; otherwise the G rule of the alert topology (if not yet in the library) and an S rule (if a state change is visible) are synthesised and certified, and certified rules enter the library. A rule acts only from the time it was learned. When the alert topology has fewer than 300 training records, its noise is calibrated online (Section IV-G) and the rule acts only after calibration. Labels are used only afterwards: per attack instance we record whether the detectors fired, whether the library fired through rules learned *before* the attack started (strict accounting), and the first firing times; per benign record (more than 60 s from any attack window) we record false alarms.

G. Noise calibration, persistence and online per-topology calibration

A certificate covers physical variation, not telemetry noise. Noise is calibrated on the attack-free training day per measurement, with a floor per measurement kind (the median over the kind of the per-measurement 99.9th percentile), because a line idle in training would otherwise get a margin of zero. Where the loop meets a topology with too few training records, the first 300 s of records in that topology on which no detector alarms serve as calibration data, without labels; the rule acts afterwards. Raising a margin cannot break a certificate. A persistence filter (fire on L consecutive records) is reported as an ablation and is part of the partially observed configuration ($L = 3$).

H. Partial observability and on-demand certification

On the 178-bus grid 47 of 122 loads, 47 of 123 DER units, 47 of 114 storage units and 96 of 360 switches have no telemetry. The invariant becomes a state-estimation residual. Per topology, a weighted least-squares estimate uses the 559 monitored measurements (load P and Q included), pseudo-measurements for unmonitored load P and Q built from monitored loads of the same profile, and a weak prior; one Gauss-Newton correction against the full AC model follows, because the linear estimate alone hides small falsifications (the maximum residual on synthesised benign states fell from 12 to 0.06 noise standard deviations with the correction [wp2b_summary.md]). Load reactive power had to be a free parameter: with Q fixed at nominal the residuals on real telemetry reached 72 to 86 sigma [decision_log.md]. De-energised buses are excluded. The SE rule of topology T fires when some monitored measurement's normalised residual exceeds its margin $m_i = 1.5 \times \max(\text{noise}_i, \text{bound}_i(T))$; the bound is found by the search layer (sampling plus differential evolution on $\max_i |r_i|/m_i$), with CEGIS raising margins on counterexamples. Because the residual involves AC power flows, this certificate is search-based. Topologies are certified **on demand**: the first time one appears in telemetry, its rule is certified and acts from then on; the compute time is reported as latency.

V. EXPERIMENTAL SETUP

A. Dataset

Sherlock [14] (Zenodo record 18467070) provides, per scenario, an attack-free training day and a test day with attacks and benign operator procedures, as IEC 104 captures, IPAL process states at 1 Hz from the control-centre vantage point, ground-truth physical snapshots and labels. We use the process states as telemetry and the snapshots only to validate reconstructions and to measure attack anatomy. Table I summarises the three scenarios. Attack types: **control-and-freeze** (telemetry of an area frozen while the grid changes), **drift-off** (voltage readings

drifted away from the truth), **industroyer** (valid switching and disconnection commands) and **arp-spoof-dos** (a network attack; excluded from the process-attack evaluation by a decision taken before the experiments, and reported separately).

Table I. Scenarios.

	01-Basic	**02-Semiurban**	**03-Rural**
Base grid	CIGRE MV, all DER	SimBench 1-MV-semiurb–2-sw plus 56 DER-and-storage transformer groups	SimBench 1-MV-rural–2-sw
Buses / transformers / switches	15 / 2 / 32	178 / 58 / 360	99 / 2 / 204
Telemetry	fully monitored, 1 Hz	about 40 % unmonitored; every 5th record used	about 40 % unmonitored; every 5th record used
Training day	yes	yes	**none**
Test attacks (CFC / drift-off / industroyer / arp)	5 / 4 / 7 / 2	5 / 9 / 9 / 6	5 / 9 / 8 / 6

B. Grid reconstruction

The download contains no grid model. Each grid was reconstructed from data and validated against the ground-truth snapshots [sherlock_model_check.md, semiurban_identify.md, semiurban_model_check_test.md, rural_identify.md, rural_model_check_test.md]:

- **01-Basic**: pandapower's CIGRE MV network with all DER; the 24 line switches were mapped from the data (an open mapped switch carries less than 1 W at its line end in 172 of 172 cases); transformer impedances and iron losses were fitted to recorded losses; the tap changer was identified as HV side, 1.5 % per step, on held-out snapshots (voltage error 2.68e-4 pu mean with the identified tap against 1.23e-2 pu ignoring it [sherlock_taps.md]).
- **02-Semiurban**: SimBench 1-MV-semiurb–2-sw plus 56 RTU groups, each a new bus with a 0.63 MVA transformer, a DER unit and a storage unit, found from the IEC 104 point map and bus power balances (maximum balance error 5.69e-5 MW).
- **03-Rural**: exactly SimBench 1-MV-rural–2-sw; fitted transformers match SimBench (vk 11.99 % against 12.00 %).

Table II. Reconstruction accuracy against ground truth (100 random snapshots; mean / maximum absolute error).

Grid	**Bus voltage, pu**	**Line P, MW**	**Transformer P, MW**
01-Basic, published parameters	1.23e-3 / 3.25e-3	2.61e-5 / 1.08e-3	3.38e-2 / 7.20e-2
01-Basic, after identification	2.60e-4 / 1.06e-3	3.69e-6 / 1.98e-4	3.83e-4 / 5.00e-4
02-Semiurban (test day)	2.75e-4 / 1.03e-3	6.74e-6 / 2.07e-4	3.18e-6 / 3.81e-4
03-Rural (test day)	2.86e-4 / 1.01e-3	5.23e-6 / 2.92e-4	6.29e-5 / 3.45e-4

The measurement noise floor, the largest difference between the same voltage recorded twice (bus and line end), is 5.0e-4 pu. All three reconstructions reproduce the recorded voltages at that level.

C. Detectors and calibration

- **D1** (01-Basic): AC power flow from the metered injections, switch and tap states; residual = largest absolute difference against reported bus voltages, line P and transformer P; alarm when a residual stays above $\theta = 2\times$ its training 99.9th percentile for L records, $L = 1.5\times$ the longest training run above θ (here θ_{vm} = 2.50e-3 pu, L = 9) [physics_residual_sherlock.md].
- **SE-BDD** (02-Semiurban, 03-Rural): the state estimator of Section IV-H; score = largest normalised residual; alarm when the score exceeds $2\times$ the training 99.9th percentile (θ = 170.45) for L = 3 records [se_bdd_semiurban.md].
- **ML**: an MLP autoencoder (64-12-64) and an isolation forest (200 trees), each on raw telemetry or on physics-informed features (D1 or SE residual vectors plus rates of change), fit on the first 75 % of the training day, threshold at the 99.95th percentile of scores on the last 25 %, with persistence [ml_detectors_*.md].

All thresholds and noise levels come from the training day only. On 03-Rural, which has no training day, nothing is calibrated on Rural data: envelope, thresholds and noise floors are transferred (Section VI-D).

D. Metrics and statistics

Per attack instance: detected if any alarm falls in [start, recovery]; first-alarm delay. Per benign record: false-alarm rate and false-alarm episodes per benign hour. Wilson 95 % intervals for detection rates, exact Poisson (Garwood) intervals for episode rates, and the exact McNemar test (two-sided binomial on discordant instances) for paired comparisons between methods on the same instances [statistics.md]. Seeds: 7 for all main results, 8 and 9 for robustness. Every script runs in under 10 minutes on a 4-core CPU; package versions are in `reports/env.txt` (pandapower 3.5.6, z3-solver 5.4.1, simbench 1.6.3).

VI. RESULTS

A. The oracle separates sound rules from mined ones

Three hand-written rules test the oracle before any synthesis (Table III). R1 is a naive voltage threshold mined from a narrow training envelope; R2 a context rule (line loading above the training 99th percentile *and* downstream load below the 1st percentile); R3 the active-power balance of the grid with a margin calibrated by search.

Table III. Oracle verdicts, robust exact layer with the linearised AC model [oracle_exact_linac.md, oracle_search.md]. All verdicts agree with the AC search layer.

Rule	**Reference CIGRE MV (16 configurations)**	**Sherlock 01-Basic (336 configurations)**
R1: $v_6 < 0.9333$ pu	refuted (253 of 9,671 benign samples fire it)	refuted (204 of 9,999; 281 of 336 configurations SAT)
R2: $\text{loading}_0 > 83.7$ % and downstream load < 0.919 MW	refuted	certified (336 UNSAT, 104 s total)
R3: $\lvert\sum P_{hv} - \sum P_{load} - \sum P_{sto} + \sum P_{der}\rvert > m$	certified	certified (336 UNSAT, 13 s total)

Two observations carry over to synthesis. First, calibrating R3's margin by sampling is unsafe: 2,000 samples gave a maximum benign residual of 0.3758 MW, and differential evolution found 0.6048 MW in the Sherlock envelope; the margin is therefore 1.5× the search maximum [rules_sherlock.json]. Second, **a safe threshold mined from history is a useless one**: CEGIS on R1 lowers τ from 0.93327 to 0.89998 pu, below the 0.90 pu operating limit, leaving 0.0000 of the original firing band [cegis_demo_linac_sherlock.md]. Synthesis must therefore produce context and invariant rules, not thresholds (Fig. 3).

 Fig. 3

Fig. 3

Fig. 3. Counterexample-guided refinement of the mined threshold rule R1 on both grids: the threshold is driven below the 0.90 pu operating limit, leaving no firing region [cegis_demo_linac.md, cegis_demo_linac_sherlock.md].

B. The certified rules

This section lists the rules the method produced on 01-Basic, in the form the library stores them. Measurement names are those of the telemetry: `vm_pu.b` bus voltage, `p_from.l` line active power at the from end, `load_p.i`, `sgen_p.i`, `storage_p.i` injections in MW, `line_in.l` the line's in-service state.

Consolidation library (10 rules, 8 G and 2 S) [consolidation_sherlock.md, library_sherlock.json]:

#	Rule (as stored)	Family	Learned from	Reading
1	`line_in.2 = 0 ∧ line_in.3 = 0 ∧ load_p.2 < 0.0043 ∧ load_p.3 < 0.0073 ∧ load_p.4 < 0.0055 ∧ sgen_p.1 < 0.0002 ∧ sgen_p.2 < 0.0003 ∧ sgen_p.3 < 0.0003 ∧ sgen_p.9 < 0.0003 ∧ storage_p.0 < ...`	S	episode at 4,293 s (industroyer, two switches opened)	the zone behind lines 2 and 3 is islanded and dark; no benign state in the envelope has it
2	`G@ties=none;out=none;tap=0,0`: 58 clauses, one per bus voltage and active line flow and sign, e.g. `vm_pu.10 − (0.98706 + Σ_i a_i·load_p.i − Σ_j a_j·sgen_p.j − ...) > 0.00722` with the guard that all lines are in and taps at 0	G	4,505 s (drift-off)	a reported network quantity is inconsistent with the metered injections in the base topology
3	`G@ties=S2;out=L4;tap=0,0` (56 clauses)	G	14,729 s	same, after a benign maintenance

#	Rule (as stored)	Family	Learned from	Reading
				reconfiguration (line 4 out, tie S2 closed)
4	`line_in.3 = 0 ∧ load_p.3 < 0.0073 ∧ sgen_p.2 < 0.0003 ∧ sgen_p.9 < 0.0003`	S	17,737 s (industroyer)	line 3 out with its load and both DER dark
5	`G@ties=S1+S2;out=L4+L11;tap=0,0` (54 clauses)	G	19,744 s	topology with two lines out and two ties closed
6	`G@ties=S1+S2;out=L11;tap=0,0` (56 clauses), e.g. `vm_pu.1 − Σ a_i x_i > 1.00761` (31 injection terms)	G	later episode	after line 4 returns
7	`G@ties=S1+S2;out=L11;tap=-1,0` (56 clauses)	G	later episode	same topology after a benign tap change on transformer 0
8	`G@ties=S1+S2;out=L9;tap=0,0` (56 clauses)	G	later episode	
9	`G@ties=S1+S2+S3;out=L7+L9;tap=0,0` (54 clauses)	G	later episode	meshed configuration
10	`G@ties=S1+S2+S3;out=L8+L9;tap=0,0` (54 clauses)	G	later episode	meshed configuration

Fig. 4 shows the loop over the test day: when each rule was learned, which alerts were already covered, which candidates the oracle refused, and when the detectors and the library first fired on each attack.



Fig. 4

Fig. 4. The self-consolidation loop on the 01-Basic test day. Top: attack windows by type, benign operator procedures, first alarms of the detectors and of the library (rules learned before the attack only), alert episodes (certified G or S rule, covered by the library) and refuted candidates. Bottom: library size [consolidation_tc_sherlock.md].

Refuted candidates (the oracle found a benign state that fires them): `S-line 4 out` and `S-line 11 out` (a line out of service alone is benign maintenance), `S-sgen_p.0 to zero` (refuted by search: a DER at zero output is benign) and `S-sgen_p.6 to zero ∧ sgen_p.12 to zero` (refuted by the exact layer) [consolidation_sherlock.md, synthesis_sherlock.md]. These refutations are the point of the gate: each would have been a plausible hand-written rule and a source of false alarms.

Single-measurement invariants synthesised from labelled alerts (family P, offline synthesis study, 12 certified of 14 candidates [synthesis_sherlock.md]): `P-vm10@base`, margin 0.00722 pu (from a drift-off); `P-vm14@base`, 0.00209 pu, and `P-p11@base`, 0.00644 MW (from a control-and-freeze); `P-vm3`, `P-vm6`, `P-vm2@ties=S1+S2;out=L4+L11`, margins 0.00712, 0.00732, 0.00551 pu; `P-p1@ties=S1+S2;out=L4+L11`, 0.153 MW and, after a second alert, 0.0915 MW; `S-load_p.9 ∧ load_p.17 to zero`; `S-load_p.16 to zero`. The certified rules fired on at most 0.0003 of benign test records each.

Partially observed grid (02-Semiurban). The library holds one SE rule per topology, 13 by the end of the test day, each of the form "some monitored residual $|r_i|$ exceeds m_i", with per-kind noise floors in units of sigma: load P 0.16, load Q 0.34, line P 1.64, DER P 0.16, storage P 0.16, transformer P 0.47, bus voltage 2.50 [semiurban_consolidation_semiurban.md]. The topology labels are readable, for example `out=L20+L113+L114+L115+L116+L117+L119+L120` for the initial configuration and `out=L4+L7+L21+L42+L53+L109+L115+L116+L117` for the last.

C. Main comparison on the recorded test days


Fig. 5

Fig. 5

Fig. 5. Instances detected per attack type and method on the recorded test days and on the additional day of 02-Semiurban, with false alarms per benign record [comparison.md, comparison_split.md].


Fig. 6

Fig. 6

Fig. 6. False alarms per benign record on the three grids, log scale. Classic physics detectors and the certified library sit two to three orders of magnitude below machine-learning detectors on raw telemetry; on 03-Rural the library needs local noise calibration [comparison.md].

Table IV. Detection per attack type (instances detected / instances), mean first-alarm delay over detected process attacks, and false alarms per benign record [comparison.md].

Grid	Method	CFC	drift-off	industroyer	delay s	FA / record
01-Basic	D1 residual test	5/5	3/4	2/7	78	0.00059
01-Basic	AE, raw telemetry	5/5	4/4	7/7	1	**0.99988**
01-Basic	AE, physics-informed	4/5	3/4	2/7	92	0.00100
01-Basic	Detectors D1 or AE (alert source)	5/5	3/4	2/7	77	0.00109
01-Basic	Certified library, strict	4/5	**4/4**	2/7	102	0.00124
01-Basic	Combined: detectors or library	5/5	4/4	**3/7**	**68**	0.00171
02-Semiurban	SE-BDD	3/5	4/9	0/9	261	0.00000

Grid	Method	CFC	drift-off	industroyer	delay s	FA / record
02-Semiurban	AE, raw telemetry	5/5	9/9	9/9	19	**0.85948**
02-Semiurban	AE, SE-residual features	4/5	6/9	7/9	121	0.14888
02-Semiurban	Certified library, single record	5/5	7/9	7/9	123	0.01586
02-Semiurban	Certified library, L = 3	**4/5**	**6/9**	0/9	**124**	0.00401
02-Semiurban	Combined: SE-BDD or library (L = 3)	4/5	6/9	0/9	124	0.00401

On the fully observed grid the library alone, under strict accounting, catches every drift-off including the one D1 misses, and the combination adds one industroyer instance and cuts the mean delay from 77 to 68 s. On the partially observed grid the library with persistence detects more than SE-BDD at 0.004 false alarms per record, and all 13 test topologies (12 unseen in training) were certified on demand with a median latency of 5 s and a maximum of 10 s [semiurban_consolidation_semiurban.md]. Industroyer, which changes the grid through valid commands and keeps the telemetry consistent, is caught by physics-consistency rules only through switching transients (single-record mode, 7/9 at 0.016 false alarms) or through state rules when the resulting state is outside the envelope.

D. Zero-training transfer to 03-Rural

03-Rural has no training day (first attack at 39 minutes). Nothing was calibrated on Rural data: the active-power envelope comes from the public SimBench profiles widened by 0.05 (covers 100 % of benign monitored load P); reactive-power bounds and the SE-BDD threshold are transferred from 02-Semiurban [rural_envelope_rural.md, se_bdd_rural.md].

- SE-BDD with the transferred threshold: drift-off 5/9, control-and-freeze 0/5, industroyer 0/8, false alarms 0.00017 per record. **The detector’s global threshold transfers.**
- Certified library with transferred per-kind noise floors: detects nearly everything but at 0.723 false alarms per record with persistence, because Rural’s transformer flow residuals reach 19 sigma against a transferred floor of 0.47. **Per-measurement noise does not transfer.** With the transferred global threshold as minimum margin (a decision taken after this failure, and reported as such) the flood disappears, but the library adds nothing over SE-BDD.
- Calibrating the library’s noise on the first 30 minutes of the test day (attack free, excluded from evaluation) gives control-and-freeze 5/5, drift-off 8/9, industroyer 4/8 at 0.184 false alarms with persistence [semiurban_consolidation_cal30_rural.md]: 30 minutes do not cover a day’s operating range.

The physics certificate itself needs no local data; the library’s noise term needs a representative benign period on each grid.

E. An additional, independently calibrated test day

Sherlock has one test day per scenario. For a second, independent test, each recorded training day (12 h, attack free) of 01-Basic and 02-Semiurban is split at its midpoint; every calibration (envelope, D1, SE-BDD, ML, noise) is redone on the first half, and 21 process attacks per grid are injected into the second half, modelled on the measured anatomy of the real ones [attack_anatomy.md, drift_magnitude.md, split_day.md]: drift-off ramps whose final offset and bus count are drawn from the real instances of the same grid (bimodal: 0.036 to 0.39 pu on 1 to 6 buses, or about 0.0015 pu), control-and-freeze on an area of 3 or 10 buses, often with a line opened behind the freeze, and industroyer as a line opened or generators disconnected. Physical changes enter as an AC power-flow delta on the nearest ground-truth snapshot, so recorded noise and model mismatch are kept. A first model with guessed drift magnitudes (0.005 to 0.05 pu), run before the real ones were measured, is reported as a weak-attack sensitivity case; the measured model is the main result. The choice follows the anatomy, not the detection numbers (the weak model favoured the library more).

Table V. Split day, measured attack model [comparison_split.md].

Grid	Method	CFC	drift-off	industroyer	FA / record
01-Basic	D1	7/7	4/7	0/7	0.00059
01-Basic	Certified library, strict	4/7	2/7	0/7	0.01432
01-Basic	Library with per-topology calibration, strict	4/7	2/7	0/7	0.00346
01-Basic	Combined (per-topology calibration)	7/7	4/7	0/7	0.00425
02-Semiurban	SE-BDD	1/7	1/7	0/7	0.00000
02-Semiurban	AE, SE-residual features	7/7	3/7	0/7	0.05913
02-Semiurban	Certified library, L = 3	**7/7**	**4/7**	0/7	**0.00000**

Every visible drift-off is caught; the three misses per grid are the near-zero instances (about 0.0015 pu). On the partially observed grid the library detects 11 of 21 at zero false alarms where SE-BDD detects 2 (9 discordant against 0, $p = 0.004$): one threshold at 138 sigma hides what per-measurement margins expose. On the fully observed grid D1 already catches every visible attack, and the library's false alarms came from benign records in a topology rare in the first half, where the real grid deviates from the reconstructed model by more than the training noise; online per-topology calibration (Section IV-G) cuts them from 0.0143 to 0.0035 per record with detection unchanged, and the remainder are runs of 2 to 6 records around benign operator actions. Industroyer is 0 for every method: injected switching is physically consistent by construction, and the real-day catches came partly from transients a power-flow delta does not reproduce. Finally, a benign command moved a transformer to a tap position unseen in the first half, after which synthesis stopped for two hours; padding the tap range by one step, with bounds re-validated, restored certification (control-and-freeze 6/7) [consolidation_sherlock_split_v1_tappad.md].

F. Uncertainty and paired tests

Table VI. Pooled over process attacks [statistics.md, statistics_split.md]. Wilson 95 % intervals; exact McNemar test for combined against classic (D1 or SE-BDD).

	Recorded test days (61 instances)	**Recorded days plus split day (103 instances)**
Classic detector	22/61 = 0.36 [0.25, 0.49]	35/103 = 0.34 [0.26, 0.44]
Certified library	25/61 = 0.41 [0.30, 0.54]	42/103 = 0.41 [0.32, 0.50]
Combined	27/61 = 0.44 [0.33, 0.57]	49/103 = 0.48 [0.38, 0.57]
Only combined / only classic detects	5 / 0, p = 0.062	14 / 0, **p = 0.00012**



Fig. 7

Fig. 7. Pooled detection of process attacks with Wilson 95 % intervals and the exact McNemar test of the combined system against the classic detector [statistics.md, statistics_split.md].

The combined system never loses an instance the classic detector catches. Per grid and day, the strongest paired evidence is on the partially observed grid.

G. Robustness and ablations

- **Seeds 7, 8, 9** (oracle search, ML training): per-type detection identical on both grids; 01-Basic library false alarms 0.00115 to 0.00124 [comparison.md].
- **Persistence** on the library: L = 1, 3, 5 gives 0.00124, 0.00091, 0.00065 false alarms on 01-Basic with unchanged detection; on 02-Semiurban L = 3 removes the industroyer transients (7/9 to 0/9) and cuts false alarms from 0.01586 to 0.00401.
- **Margin factor**: at 1.25 the grid-wide rules cannot be certified on 01-Basic, because the robust bound (1.25 × oracle bound) is not below the margin, and the library is left with 3 state rules; 1.5 (default, fixed in advance) as reported; 2.0 gives the same detection at 0.00109 (01-Basic) and 0.00174 (02-Semiurban). The margin factor must exceed the bound safety factor; above that it trades false alarms for delay.
- **Per-topology online calibration** (Section IV-G) on the recorded 01-Basic day: library false alarms 0.00124 to 0.00056 per record (0.64 to 0.21 episodes per hour), detection unchanged, now below D1's 0.85 episodes per hour [comparison.md]. Fig. 8 summarises this and the Rural transfer.



Fig. 8

Fig. 8. What calibration does. (a) 01-Basic: five minutes of label-free online calibration per unfamiliar topology cuts library false alarms by a factor of 2.2 on the recorded day and 4.1 on the additional day, with detection unchanged. (b) 03-Rural: the detector threshold transfers across grids, per-measurement noise floors do not, and 30 minutes of local data are not enough [comparison.md, comparison_split.md, semiurban_consolidation_*_rural.md]. - **Model error bounds**: all verdicts of the robust exact layer agree with the AC search layer on both grids; the scenario bound of Section IV-D holds for every quantity.

H. Machine-learning detectors as a negative result

Autoencoders on raw telemetry raise alarms on 0.99988 (01-Basic), 0.85948 (02-Semiurban) and 0.43249 (03-Rural, trained on 30 minutes) of benign records, because loads, DER output and topology change through the day in ways the training day does not pin down. Physics-informed features repair this on the fully observed grid (0.00100) but not fully on the partially observed one (0.14888), whose test day has 12 topologies unseen in training. The isolation forest saturates and detects almost nothing with physics features on 01-Basic (0/5, 0/4, 0/7 at 0 false alarms). None of the ML detectors is close to the certified library's operating point.

VII. DISCUSSION

What the certification gate buys. Three things that neither detectors nor hand-written rules give on their own: durable, readable knowledge (Section VI-B is a complete list of what the system learned, each rule with the reason it holds); a guarantee against false alarms from physical variation inside a documented envelope; and refusal of plausible-but-wrong rules (a DER at zero, a line out) before they reach operations. The gain in detection over a well-calibrated classic detector is real but modest where the grid is fully observed, and large where it is not, because per-measurement margins resolve what a single global residual threshold averages away.

What transfers between grids and what does not. The physics certificate needs no local data: on 03-Rural every topology was certified on demand. A detector's global threshold transferred between two grids of the same dataset family. Per-measurement noise did not transfer, and 30 minutes of local data were not enough. In deployment this means a benign calibration period of the order of a day per grid, and the online per-topology calibration of Section IV-G for topologies that period did not cover.

The envelope matters more than the rule language. Every failure mode we met was an envelope or calibration failure: a tap position not in the envelope, a topology rare in calibration, reactive power assumed fixed, a line idle in training. Engineering ranges (one tap step beyond those observed, every topology one benign procedure away) are the right default, with bounds re-validated when the envelope grows.

Limitations and threats to validity. (1) One recorded test day per grid and 4 to 9 instances per attack type; intervals are wide, and the second test day is semi-synthetic: its attacks are injected and cannot reproduce network-level side effects (no industroyer catches for any method there). (2) The exact-layer certificate is relative to a linear model with empirical error bounds; the scenario bound is relative to a uniform sampling of the envelope, not a worst case; on the partially observed grid the certificate is search-based only. (3) The grids are reconstructions, validated at the noise floor, not the original simulator scenarios. (4) The library learns and is evaluated on the same test day in time order; the strict accounting bounds the optimism. (5) The split-day attack model was revised once, after measuring the real attacks; both versions are reported. (6) Attacks through valid commands that leave the state inside the envelope (a set-point change, a redundant command) leave no trace in a static snapshot and need state-transition or command-context rules, which this paper does not synthesise.

VIII. CONCLUSION

We presented a self-consolidating intrusion detection system for distribution grids in which detector alerts become deterministic rules, and rules become operational only after a physics oracle certifies that no benign feasible operating state fires them. On three grids reconstructed from the Sherlock dataset the oracle refuted rules mined

from history and plausible-but-wrong state rules, certified physically grounded ones in seconds, and the resulting library, replayed without labels, caught attacks the detectors missed. Pooled over 103 attack instances across recorded and independently calibrated test days, combining the library with a classic detector detected 48 % of instances against 34 %, never losing an instance (14 against 0, p = 0.00012). The main practical lesson is that the certificate covers physics, not noise, and that noise must be calibrated per grid and per topology; we showed that this can be done online, without labels, from five minutes of data. Future work includes rules over state transitions and command context for attacks that stay inside the envelope, a formal rather than empirical error bound for the linear layer, and evaluation on a second recorded day per grid as such data become public.

REPRODUCIBILITY

Code: `paper3-wp0/` in the project repository (Python, pandapower, Z3). `bash scripts/run_all.sh` reproduces the main results with seed 7; `night_runs.sh`, `rural_runs2.sh`, `split_runs.sh`, `split_tappad_runs.sh` and `wp5_runs.sh` the robustness, transfer, split-day and calibration results; `scripts/19_comparison.py` and `scripts/20_statistics.py` the tables. Every report names the command that produced it. Dataset files are not redistributed; download scripts verify checksums.

REFERENCES

[1] Electricity Information Sharing and Analysis Center (E-ISAC) and SANS ICS, “Analysis of the cyber attack on the Ukrainian power grid,” Mar. 2016.

[2] Bundesamt für Sicherheit in der Informationstechnik (BSI), “Die Lage der IT-Sicherheit in Deutschland 2024,” Bonn, Germany, Nov. 2024 (incident count as cited in [14]).

[3] R. Mitchell and I.-R. Chen, “A survey of intrusion detection techniques for cyber-physical systems,” *ACM Computing Surveys*, vol. 46, no. 4, 2014.

[4] K. Wolsing, E. Wagner, A. Saillard, and M. Henze, “IPAL: Breaking up silos of protocol-dependent and domain-specific industrial intrusion detection systems,” in *Proc. 25th Int. Symp. Research in Attacks, Intrusions and Defenses (RAID)*, Limassol, Cyprus, 2022, pp. 510–525, doi: 10.1145/3545948.3545968.

[5] A. Solar-Lezama, L. Tancau, R. Bodík, S. Seshia, and V. Saraswat, “Combinatorial sketching for finite programs,” in *Proc. ASPLOS*, 2006.

[6] L. de Moura and N. Bjørner, “Z3: An efficient SMT solver,” in *Proc. TACAS*, 2008.

[7] G. C. Calafiore and M. C. Campi, “The scenario approach to robust control design,” *IEEE Trans. Automatic Control*, vol. 51, no. 5, pp. 742–753, 2006.

[8] M. E. Baran and F. F. Wu, “Network reconfiguration in distribution systems for loss reduction and load balancing,” *IEEE Trans. Power Delivery*, vol. 4, no. 2, pp. 1401–1407, 1989.

[9] D. Hadžiosmanović, R. Sommer, E. Zambon, and P. H. Hartel, “Through the eye of the PLC: Semantic security monitoring for industrial processes,” in *Proc. 30th Annual Computer Security Applications Conf. (ACSAC)*, New Orleans, LA, USA, 2014, pp. 126–135, doi: 10.1145/2664243.2664277.

[10] C. Feng, V. R. Palleti, A. Mathur, and D. Chana, “A systematic framework to generate invariants for anomaly detection in industrial control systems,” in *Proc. Network and Distributed System Security Symp. (NDSS)*, San Diego, CA, USA, 2019.

[11] Y. Chen, C. M. Poskitt, and J. Sun, “Learning from mutants: Using code mutation to learn and monitor invariants of a cyber-physical system,” in *Proc. IEEE Symp. Security and Privacy*, 2018.

[12] Q. Lin, S. Adepu, S. Verwer, and A. Mathur, “TABOR: A graphical model-based approach for anomaly detection in industrial control systems,” in *Proc. ASIACCS*, 2018.

[13] K. Wolsing, L. Thiemt, C. van Sloun, E. Wagner, K. Wehrle, and M. Henze, “Can industrial intrusion detection be SIMPLE?,” in *Proc. 27th European Symp. Research in Computer Security (ESORICS)*, LNCS vol. 13556, Springer, 2022, pp. 574–594, doi: 10.1007/978-3-031-17143-7_28.

[14] E. Wagner, L. Bader, K. Wolsing, and M. Serror, “Sherlock: A dataset for process-aware intrusion detection research on power grid networks,” in *Proc. 15th ACM Conf. Data and Application Security and Privacy (CODASPY)*, Pittsburgh, PA, USA, 2025, doi: 10.1145/3714393.3726006.

[15] A. Abur and A. Gómez Expósito, *Power System State Estimation: Theory and Implementation*. New York: Marcel Dekker, 2004.

[16] D. I. Urbina, J. A. Giraldo, A. A. Cárdenas, N. O. Tippenhauer, J. Valente, M. Faisal, J. Ruths, R. Candell, and H. Sandberg, “Limiting the impact of stealthy attacks on industrial control systems,” in *Proc. ACM CCS*, 2016.

[17] J. Giraldo, D. Urbina, A. Cárdenas, J. Valente, M. Faisal, J. Ruths, N. O. Tippenhauer, H. Sandberg, and R. Candell, “A survey of physics-based attack detection in cyber-physical systems,” *ACM Computing Surveys*, vol. 51, no. 4, 2018.

[18] Y. Liu, P. Ning, and M. K. Reiter, “False data injection attacks against state estimation in electric power grids,” *ACM Trans. Information and System Security*, vol. 14, no. 1, 2011 (conference version: ACM CCS 2009).

[19] G. Liang, J. Zhao, F. Luo, S. R. Weller, and Z. Y. Dong, “A review of false data injection attacks against modern power systems,” *IEEE Trans. Smart Grid*, vol. 8, no. 4, pp. 1630–1638, Jul. 2017, doi: 10.1109/TSG.2015.2495133.

[20] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. IEEE ICDM*, 2008.

[21] L. Bader, M. Serror, O. Lamberts, Ö. Sen, D. van der Velde, I. Hacker, J. Filter, E. Padilla, and M. Henze, “Comprehensively analyzing the impact of cyberattacks on power grids,” in *Proc. IEEE European Symp. Security and Privacy (EuroS&P)*, Delft, The Netherlands, 2023, pp. 1065–1081.

[22] CIGRE Task Force C6.04.02, “Benchmark systems for network integration of renewable and distributed energy resources,” CIGRE Technical Brochure 575, 2014.

[23] S. Meinecke, D. Sarajlić, S. R. Drauz, A. Klettke, L.-P. Lauven, C. Rehtanz, A. Moser, and M. Braun, “SimBench: A benchmark dataset of electric power systems to compare innovative solutions based on power flow analysis,” *Energies*, vol. 13, no. 12, art. 3290, Jun. 2020, doi: 10.3390/en13123290.

[24] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, “pandapower: An open-source Python tool for convenient modeling, analysis, and optimization of electric power systems,” *IEEE Trans. Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.

[25] R. Storn and K. Price, “Differential evolution: A simple and efficient heuristic for global optimization over continuous spaces,” *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997.

[26] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” *J. American Statistical Association*, vol. 22, pp. 209–212, 1927; Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, pp. 153–157, 1947; F. Garwood, “Fiducial limits for the Poisson distribution,” *Biometrika*, vol. 28, pp. 437–442, 1936.